

Exercise 3: Stored Procedures

Scenario 1: The bank needs to process monthly interest for all savings accounts.

Question: Write a stored procedure **ProcessMonthlyInterest** that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

Code:

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
IS
BEGIN

    UPDATE Accounts
    SET Balance = Balance + (Balance * 0.01)
    WHERE AccountType = 'Savings';

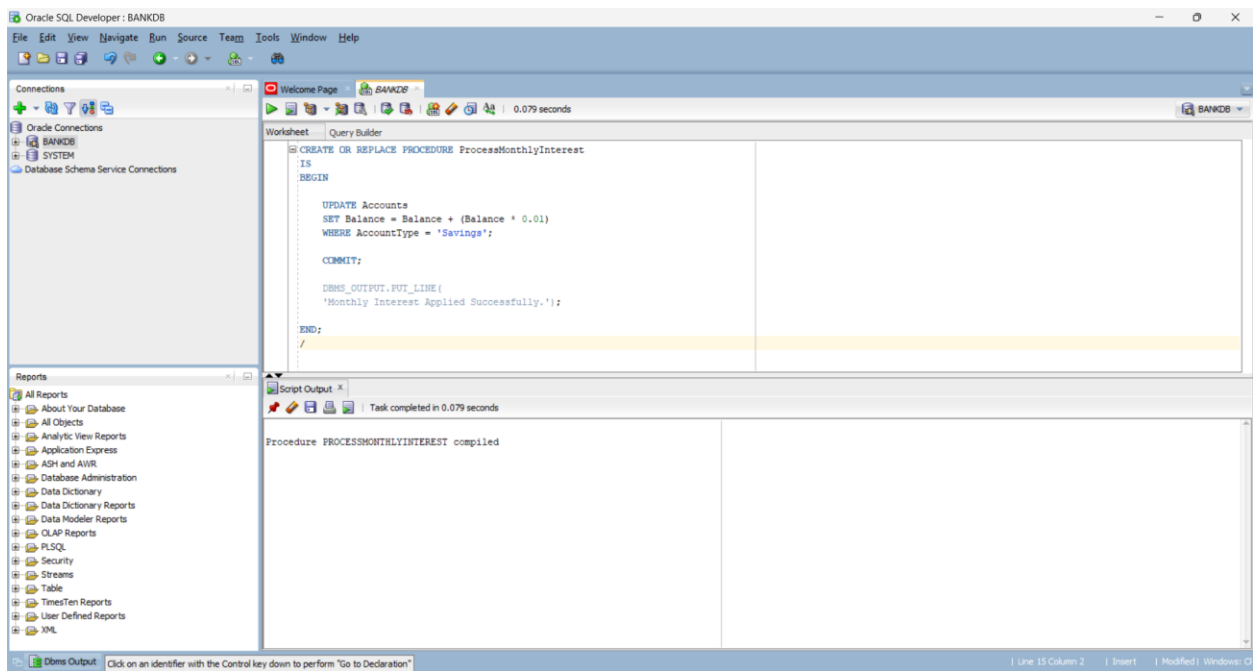
    COMMIT;

    DBMS_OUTPUT.PUT_LINE(
        'Monthly Interest Applied Successfully.');
```

END;

/

Output:



Oracle SQL Developer: BANKDB

File Edit View Navigate Run Source Team Tools Window Help

Connections

- Oracle Connections
- BANKDB
- SYSTEM
- Database Schema Service Connections

Reports

- All Reports
- About Your Database
- All Objects
- Analytic View Reports
- Application Express
- ASH and AWR
- Database Administration
- Data Dictionary
- Data Dictionary Reports
- Data Modeler Reports
- OLAP Reports
- PLSQL
- Security
- Streams
- Table
- TimesTen Reports
- User Defined Reports
- XML

Script Output

Task completed in 0.064 seconds

Procedure PROCESSMONTHLYINTEREST compiled

PL/SQL procedure successfully completed.

Line 6 Column 2 Insert Modified Windows

Oracle SQL Developer: BANKDB

File Edit View Navigate Run Source Team Tools Window Help

Connections

- Oracle Connections
- BANKDB
- SYSTEM
- Database Schema Service Connections

Reports

- All Reports
- About Your Database
- All Objects
- Analytic View Reports
- Application Express
- ASH and AWR
- Database Administration
- Data Dictionary
- Data Dictionary Reports
- Data Modeler Reports
- OLAP Reports
- PLSQL
- Security
- Streams
- Table
- TimesTen Reports
- User Defined Reports
- XML

Script Output

Task completed in 0.082 seconds

Procedure PROCESSMONTHLYINTEREST compiled

PL/SQL procedure successfully completed.

ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE	LASTMODIF
1		1 Savings	505	10-JUL-26
2		2 Checking	2000	10-JUL-26

Line 1 Column 24 Insert Modified Windows

Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

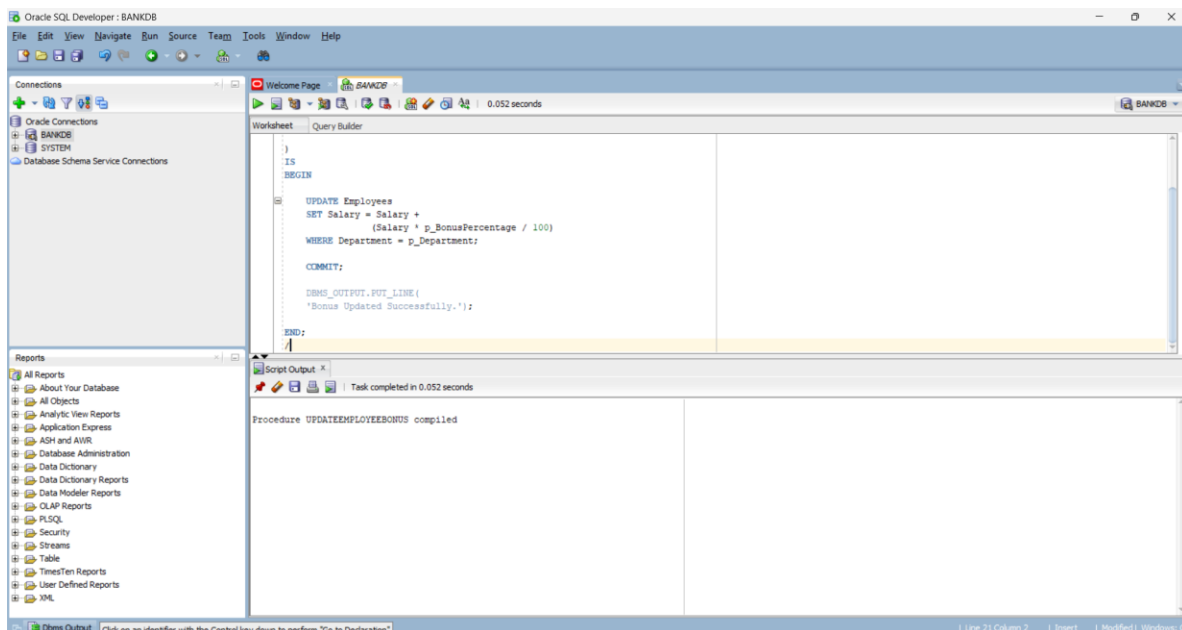
Question: Write a stored procedure **UpdateEmployeeBonus** that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

Code:

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(  
  
    p_Department IN VARCHAR2,  
    p_BonusPercentage IN NUMBER  
  
)  
IS  
BEGIN  
  
    UPDATE Employees  
    SET Salary = Salary +  
        (Salary * p_BonusPercentage / 100)  
    WHERE Department = p_Department;  
  
    COMMIT;  
  
    DBMS_OUTPUT.PUT_LINE(  
        'Bonus Updated Successfully.');
```

END;
/

Output:



Oracle SQL Developer: BANKDB

File Edit View Navigate Run Source Team Tools Window Help

Connections

- Oracle Connections
- BANKDB
- SYSTEM
- Database Schema Service Connections

Script Output

Task completed in 0.037 seconds

Procedure UPGATEEMPLOYEEBONUS compiled

PL/SQL procedure successfully completed.

Script Output

Click on an identifier with the Control key down to perform "Go to Declaration"

Line 6 Column 2 Insert Modified Windows

Oracle SQL Developer: BANKDB

File Edit View Navigate Run Source Team Tools Window Help

Script Output

Task completed in 0.044 seconds

Procedure UPGATEEMPLOYEEBONUS compiled

PL/SQL procedure successfully completed.

EMPLOYEEID	NAME	POSITION	SALARY	DEPARTMENT
1	Alice Johnson	Manager	84700	HR
2	Bob Brown	Developer	60000	IT

Script Output

Click on an identifier with the Control key down to perform "Go to Declaration"

Line 1 Column 25 Insert Modified Windows

Scenario 3: Customers should be able to transfer funds between their accounts.

Question: Write a stored procedure **TransferFunds** that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

Code:

```
CREATE OR REPLACE PROCEDURE TransferFunds(  
  
    p_FromAccount IN NUMBER,  
    p_ToAccount IN NUMBER,  
    p_Amount IN NUMBER  
  
)  
IS  
  
    v_Balance NUMBER;  
  
BEGIN  
  
    SELECT Balance  
    INTO v_Balance  
    FROM Accounts  
    WHERE AccountID = p_FromAccount;  
  
    IF v_Balance < p_Amount THEN  
  
        RAISE_APPLICATION_ERROR(  
            -20003,  
            'Insufficient Balance.');  
    END IF;  
  
    UPDATE Accounts  
    SET Balance = Balance - p_Amount  
    WHERE AccountID = p_FromAccount;  
  
    UPDATE Accounts  
    SET Balance = Balance + p_Amount  
    WHERE AccountID = p_ToAccount;  
  
    COMMIT;  
  
    DBMS_OUTPUT.PUT_LINE(  

```

```

        'Funds Transferred Successfully.');
```

EXCEPTION

```

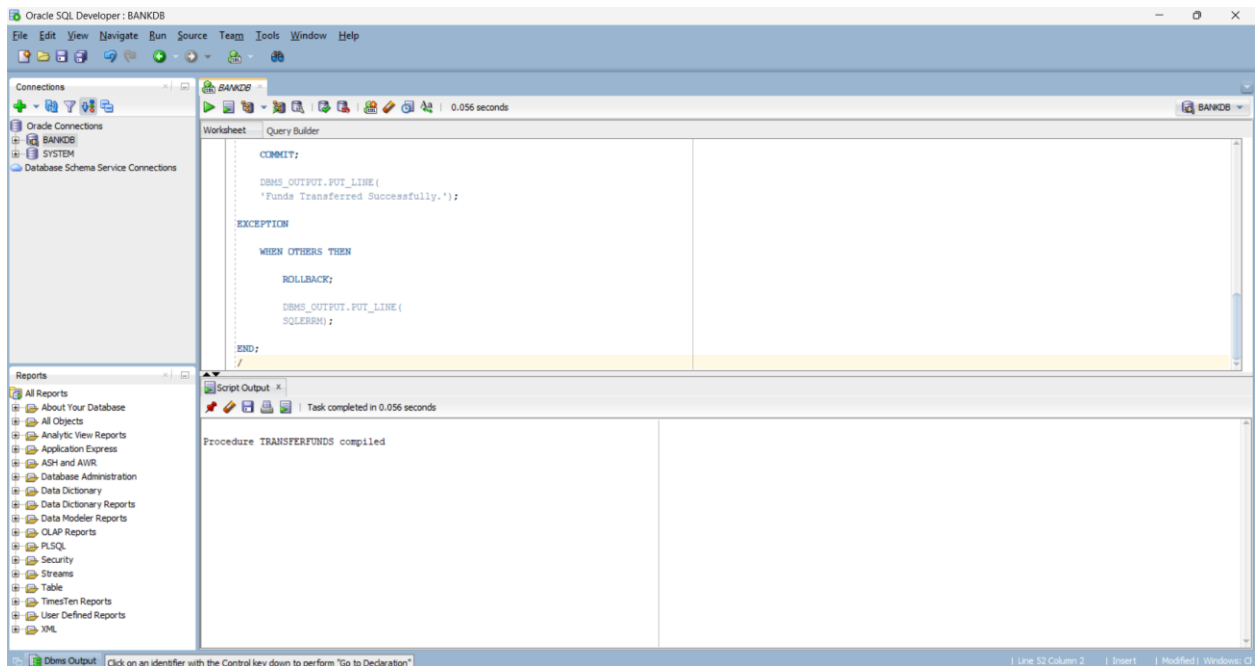
        WHEN OTHERS THEN

            ROLLBACK;

            DBMS_OUTPUT.PUT_LINE(
                SQLERRM);

END;
/
```

Output:



Oracle SQL Developer: BANKDB

File Edit View Navigate Run Source Text Tools Window Help

Connections

- Oracle Connections
 - BANKDB
 - SYSTEM
- Database Schema Service Connections

Script Output

Task completed in 0.039 seconds

Procedure TRANSFERFUNDS compiled

PL/SQL procedure successfully completed.

Script Output

```
BEGIN
  TransferFunds(1,2,500);
END;
```

Oracle SQL Developer: BANKDB

File Edit View Navigate Run Source Text Tools Window Help

Connections

- Oracle Connections
 - BANKDB
 - SYSTEM
- Database Schema Service Connections

Script Output

Query Result: All Rows Fetched: 2 in 0.007 seconds

ACCOUNTID	BALANCE
1	5
2	2500

Script Output

```
SELECT AccountID, Balance
FROM Accounts;
```